

Cs615 final term Solved

Planning process task

1. Scope planning
2. Scope definition
3. Activity definition
4. Activity Sequence
5. Activity duration estimating
6. Resource planning
7. Cost estimation

Project planning objective

Resource

Cost, and

Schedule

Project planning key tasks

1. Set goal and scope
2. Select lifecycle
3. Set org team

4. Start team selection
5. Determine risk
6. Create wbs
7. Identify task
8. Estimate size
9. Estimate effort

Primary planning steps in the planning phase

1. Project plan
2. Resource plan
3. Financial plan
4. Quality plan
5. Risk plan
6. Acceptance plan
7. Communication plan

Input to SDP (software development plan)

1. Other planning outputs
2. Historical info
3. Organizational policies
4. Constraint
5. Assumptions

Tools and techniques of SDP

1. Project planning methodology
2. Stakeholders' skills knowledge and experience
3. Project management info sys (PMIS)
4. Earned value management (EVM)

Output from SDP

1. Project plan

Input to SDP execution

1. Project plan
2. Support details
3. Organizational policies
4. Prevention action
 - Preventive action is anything that reduces the probability of potential consequences of project risk event
5. Corrective action
 - **Corrective Action:** Fixing things that have gone wrong to get a project back on track

Output from SDP execution

1. Work results
2. Change request

Element of software project plan

1. Scope planning
2. Objectives / business requirement
3. Technical approach
4. Contractual aspect
 - a. Cost plus
 - b. Fixed price (types of contract)
 - Cost –Plus

Cost-plus is a contractual relationship where the developer is paid for the cost of the service provided and in addition is allowed an agreed profit margin

- 2. Fixed Price

A fixed price contract is a commitment by the developer to provide an agreed

product or service for an agreed fee, within an agreed schedule

Process models for software projects

1. Waterfall model

- This is the traditional life cycle model. It assumes that all phases in a software project are carried out sequentially and that each phase is completed before the next is taken up.

2. Prototyping model

- The Prototyping Model is an iterative cycle that involves gathering customer requirements, producing a prototype based on those specifications, and validating the prototype with customer feedback. This process is repeated, with each iteration building on the previous prototype, until the customer is satisfied with the final product.

3. Incremental Model:

- A mix of Waterfall and Prototyping models. The project is divided into smaller, manageable chunks called

"builds". Each build adds new features to the software, and the entire project is planned around these builds

3. Spiral model

- Another evolutionary life cycle model that combines the linear nature of the Waterfall model and the iterative nature of the Prototyping model. Dividing the project life cycle into phases. Each phase is repeated in a cycle of iterations.

4. Rad (Rapid Application Development) model

- A Process defines the overall processes that an organization needs to follow to manage a project efficiently. A process model is flexible and does not follow any rigid rules of implementation. It is an incremental model that emphasizes relatively short development cycles (60, 90 days)

Drawbacks of waterfall model

- There cannot be a sudden crossover from one phase to the next.

- Another reason for the unpopularity of the Waterfall model is that it requires too much effort for stringent documentation-in every phase
- No working model is developed until the end. Client feedback is only considered after the project is complete
- Finally, the Waterfall model causes, as a "blocking state".

Phases of the Waterfall Model

- Analysis
- Design
- Coding
- Testing
- Implementation & Design

You should use Prototyping model when:

- The client is not clear about the requirements of the proposed system or
- When there is a conflict in client requirements.

Types of prototype method

1. Rapid prototyping
2. Reusable
3. Modular

1. Rapid Prototyping:

- Quick creation of a prototype to test ideas or functionality
- Emphasizes speed and simplicity over perfection

2. Reusable Prototyping:

- Create a prototype that can be used as a foundation for future development
- Design with reusability in mind to save time and resources

3. Modular Prototyping:

- Break down the project into smaller, independent modules
- Develop and test each module separately before combining them

Types of organizational system

1. Project based org
2. Non-proj based org

Organizational Cultures and Style

Shared values,

- Norms,
- Beliefs
- Expectations
- Policies and
- Procedures

Traditional Structures of Business Organization

- **Functional Structure**

A type of business structure that organizes a company into different departments based on area of expertise. functional structures when the organization is small, geographically centralized, and provides few goods and services.

- **Projectized Structure**

is designed or setup in a way that the project manager is on top of the hierarchy and has full power in any decision involved in the project.

when the organization is relatively large, geographically dispersed, and/or produces wide range of goods/services

- **Matrix Structure**

A matrix organization is a company structure where team reports to multiple leaders

Matrix organizations are a blend of functional and projectized characteristic

Input to organizational planning

1. Proj interface: 3 categories

- **Organizational interfaces**—formal and informal reporting relationships

among different organizational units.

Technical interfaces—formal and informal reporting relationships among different technical disciplines.

Interpersonal interfaces—formal and informal reporting relationships

among different individuals working on the project.

2. Staffing requirements

3. Constraint

Tools and technique for organizational planning

1. Templates
2. Human resources practice
3. Organizational theory
4. Stakeholders analysis

Output of organizational planning

1. Roles and responsibility assignment
2. Staffing management plan
3. Org charts

Microsoft proj task type's

1. General
2. Milestone
3. Recurring

Software proj type risk

1. Development process risk
2. Product-related risk

Pert and cpm common thing

1. Activities
2. Nodes

3. Critical path

4. Network

Network scheduling technique

1. Pert

2. Cpm

Guidelines for creating software proj schedule

1. Classification

2. Interdependence

3. Time and effort allocation

4. Validation criteria

5. Defined responsibilities and outputs

Create schedule with these proj management tools

1. Gantt chart

2. Network scheduling technique

Outputs from staff acquisition

1. Project staff assigned

2. Project team directory

SOW Template

- **Scope of work:** Describe the work to be done in detail.
- **Location of Work:** Describe where the work must be performed.
- **Period of Performance:** Specify when the work is expected to start and end,

Estimation of following factors are essential:

- Cost,
- Effort,
- Risks
- Resources

Estimation gives you a fair idea of the size of the project

Decomposition Techniques

Software project estimation is a form of problem solving, and in most cases, the

problem to be solved (i.e., developing a cost and effort estimate for a software

project) is too complex to be considered in one piece. For this reason, we

decompose the problem, re-characterizing it as a set of smaller (and hopefully, more manageable) problems.

Four different approaches to the sizing problem

- "Fuzzy logic" sizing
- Function point sizing
- Standard component sizing:
- Change sizing:

Inputs to team development

1. Project staff
2. Proj plan
3. Staffing management plan
4. Performance report
5. External feedback

Software proj type risk

1. Product related
2. , development process risk

Cocomo level

1. Basic
2. Intermediate
3. Advance

Wbs(Work Breakdown Structure) tools

- Program Evaluation and Review Techniques (PERT)
- Critical Path Method (CPM)
- Timeline charts
- Gantt charts

What are the main techniques to estimate effort?

- a) Source lines of Code (SLOC)
- b) Function Point (FP)
- c) Constructive Cost Model (COCOMO)

Delphi Technique

This technique is based on soft skills and relies more on human factors, such as collecting information during group discussions.

SLOC- Technique

At the beginning of a software project, it is important to determine the size of the project. The project size helps determine the resources, effort, and duration of the pro. It is defined as the source lines of code that are delivered as part of the product. The effort spent on creating the source lines of code is expressed in relation to thousand lines of code (KLOC).

FP Technique

The FP technique is a direct indicator of the functionality of a software application from the user's perspective. This is the most popular technique used to estimate the size of a software project. You use the FP technique to estimate the total size of a project. The total size of a project is estimated as a single FP value. After calculating the total size of a project in FP, you divide the total FP into the different phases of the SDLC.

Advantages of FP Technique

Total size of the project expressed total FP()

FP is independent language Ó)

FP is independent technologyÔ)

FP is independent methodologyÕ)

FP estimate any kind of project Ö)

if we use FP when we want to changing in the middle of the project

Disadvantages of FP)

It is time consuming method Ò)

Less research data is available on FPÓ)

FP provides a blurred estimationÔ)

IT need evaluation and lots of judgement

Tools for creating wbs

1. Manage risk
2. Manage cost
3. Assign work
4. Schedule and track activities

Software risk components

1. Performance risk
2. Cost risk

3. Support risk
4. Schedule risk

Factors of estimation

1. Cost
2. Effort
3. Resource and risk

technique for estimating effort

1. Sloc
2. Fp
3. Cocomo

Cocomo level

1. Basic
2. Intermediate
3. Advance

Type of wbs

1. Program wbs
2. Contract wbs

Wbs tools

1. Pert
2. Cpm
3. Gantt charts
4. Timelin charts

learning with miss khattak